AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 11 · STUDENT LESSON

Failure budgets and service objectives

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Evaluation pipelines. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of failure budgets and service objectives. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **failure budget**s and **service objective**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Evaluation pipelines. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **failure budget**s and **service objective**s using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Failure budgets and service objectives case. Separate observations, given facts and assumptions before choosing an explanation.

Service objectives define measured reliability over a stated interval. A **failure budget** is meaningful only after success and **eligible request**s are defined.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Failure budgets and service objectives. Connect each labelled input, deciding step and result.

For a fictional objective at least 98% of 200 **eligible request**s must pass all contract checks. Allowed failures are 2%*200=4. Count every eligible failure once; report excluded requests separately.

Failure budgets and service objectives — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
In the interval 200 requests are eligible and 5 fail.	For a fictional objective at least 98% of 200 eligible requests must pass all contract checks.	195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Failure budgets and service objectives. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Repair one failed request into a success in this window.

Predicted result

Prediction: 98 successes and 2 failures meet the stated window target.

Why the result follows

Reason: One recorded outcome changes while the denominator stays 100.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Failure budgets and service objectives to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Service objective	A declared measured target over a specified scope
Failure budget	The failures allowed under that target
Eligible request	A request included under the declared denominator rule

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Failure budgets and service objectives, predict a step, then compare with its explanation.

In the interval 200 requests are eligible and 5 fail.

195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

Worked explanation · Failure budgets and service objectives

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

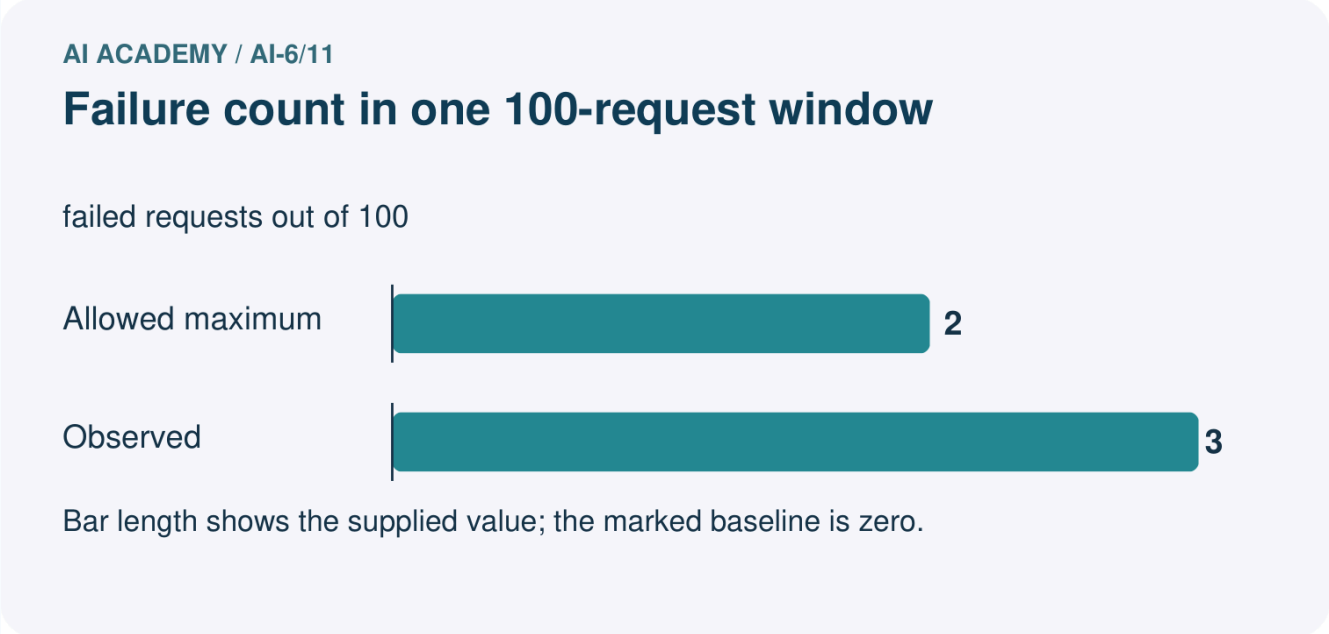
A pictorial worked case: Failure budgets and service objectives

A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.

The facts we will use

Supplied facts

Current window has 97 successes and 3 failures. Success includes a correct refusal within the agreed time limit.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/11

Failure budgets and service objectives

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define success

Use the same predeclared correctness and timing criteria.



2. Compute allowance

The target permits at most 2 failures in 100 requests.



3. Compare record

Three failures misses the target by one request.

CHANGE ONE THING

Repair one failed request into a success in this window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define success	Use the same predeclared correctness and timing criteria.
2. Compute allowance	The target permits at most 2 failures in 100 requests.
3. Compare record	Three failures misses the target by one request.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

What this example does not establish

Keep the stated limit

Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

Keep the stated limit

Read the labelled picture: Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

Keep the stated limit

Change one thing: Repair one failed request into a success in this window. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Failure budgets and service objectives claim and explain a repair using the supplied evidence.

Meeting one interval objective does not establish every future interval or zero critical failures.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Failure budgets and service objectives with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Failure budget**s and **service objective**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Failure budget**s and **service objective**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Failure budgets and service objectives. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Failure budgets and service objectives. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled **failure budget**s and **service objective**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **failure budget**s and **service objective**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Failure budgets and service objectives case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Failure budgets and service objectives — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
In the interval 200 requests are eligible and 5 fail.	For a fictional objective at least 98% of 200 eligible requests must pass all contract checks.	195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Failure budgets and service objectives, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.

By the end, I can	How I will show it
Explain the mechanism and terminology of failure budgets and service objectives	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Failure budgets and service objectives without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Three failures misses the target by one request.

1. Define success

2. Compute allowance

3. Compare record

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare record Three failures misses the target by one request. An objective needs a measurement window, denominator and success definition.

An honest refusal can be success under an agreed contract.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Repair one failed request into a success in this window. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. 98 successes and 2 failures meet the stated window target.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: 98 successes and 2 failures meet the stated window target. One recorded outcome changes while the denominator stays 100.

Repair a missing building block

Review Evaluation pipelines (ai-6/10). Goal: Explain the **mechanism** and terminology of evaluation pipelines. Short practice: Calculate A’s and B’s correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the **mechanism** and terminology of testing model-dependent behaviour.

Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Review Authentication and authorisation (ai-6/5). Goal: Explain the **mechanism** and terminology of authentication and authorisation. Short practice: Explain why Jo’s valid volunteer login does not permit editing C.

Which trusted role change allows the action, and why is a role written in the request insufficient?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Failure budgets and service objectives graphic using its labels, deciding evidence and relevant limitation.

01. Failure budgets and service objectives: the case



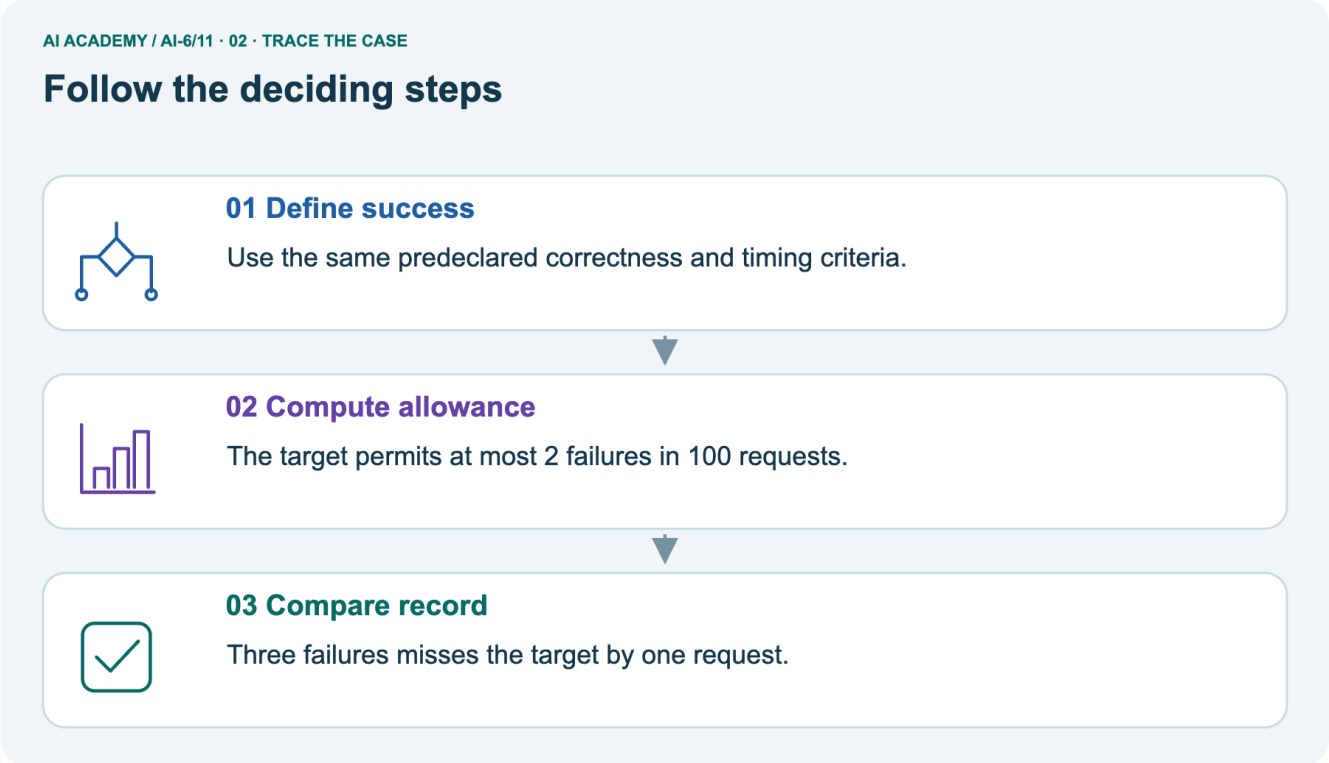
Follow the labelled picture with the matching explanation. Open the vector diagram

A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s. Supplied facts: Current window has 97 successes and 3 failures.

Success includes a correct refusal within the agreed time limit.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Define success: Use the same predeclared correctness and timing criteria. → Compute allowance: The target permits at most 2 failures in 100 requests. → Compare record: Three failures misses the target by one request.

Point to each numbered stage and explain which recorded **input** it uses.

03. Failure count in one 100-request window

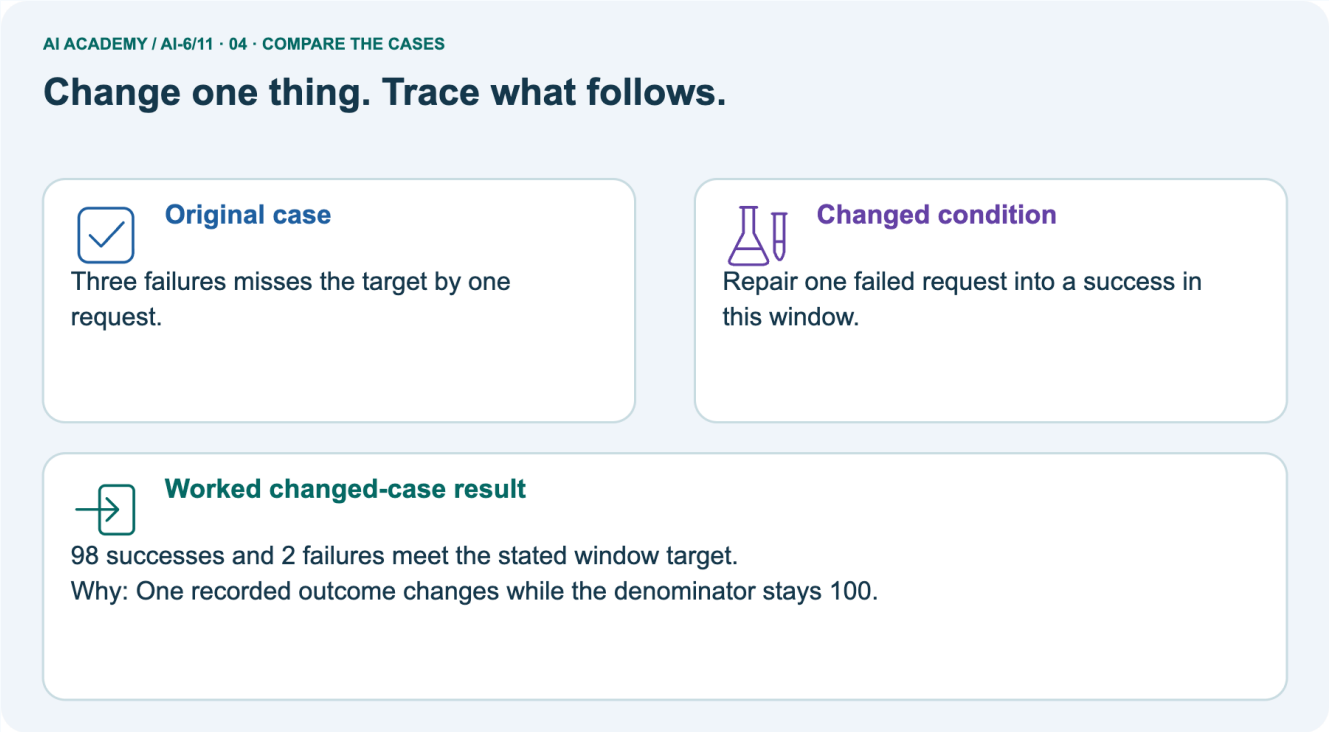


Follow the labelled picture with the matching explanation. Open the vector diagram

Failure count in one 100-request window. Units: failed requests out of 100. Allowed maximum: 2. Observed: 3.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



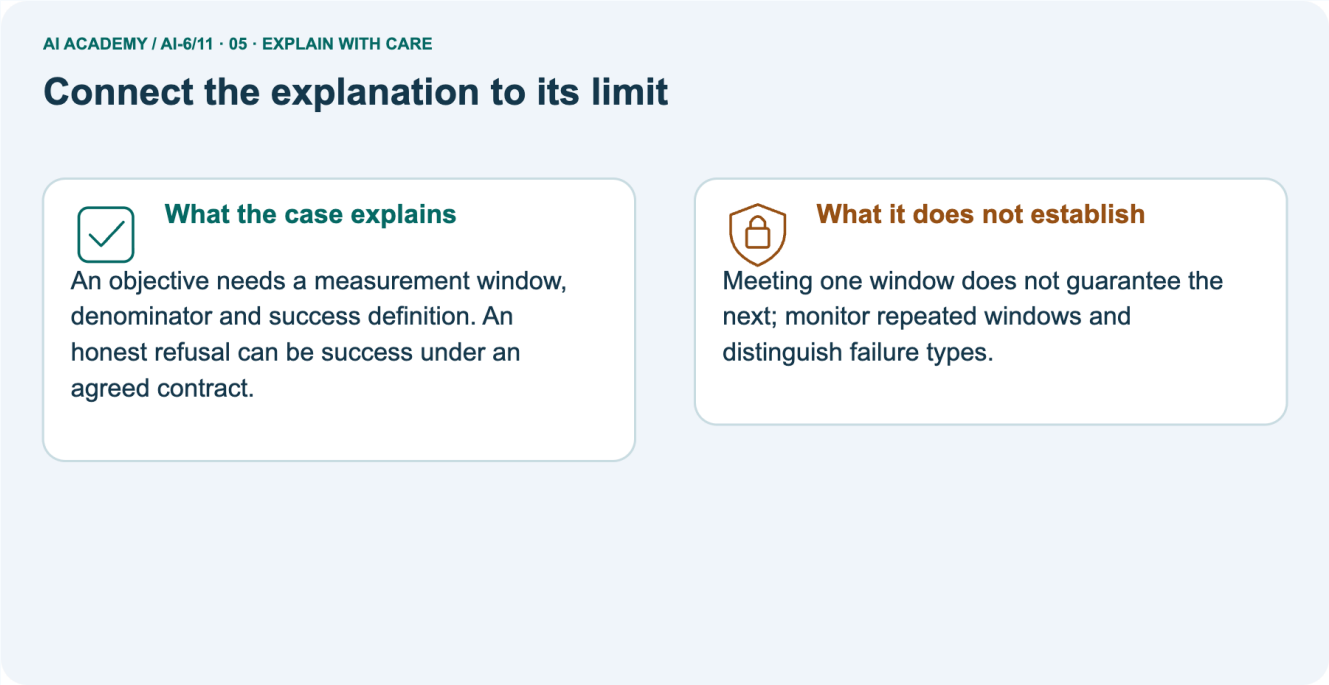
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three failures misses the target by one request. Changed condition: Repair one failed request into a success in this window. Expected result: 98 successes and 2 failures meet the stated window target.

Reason: One recorded outcome changes while the denominator stays 100.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



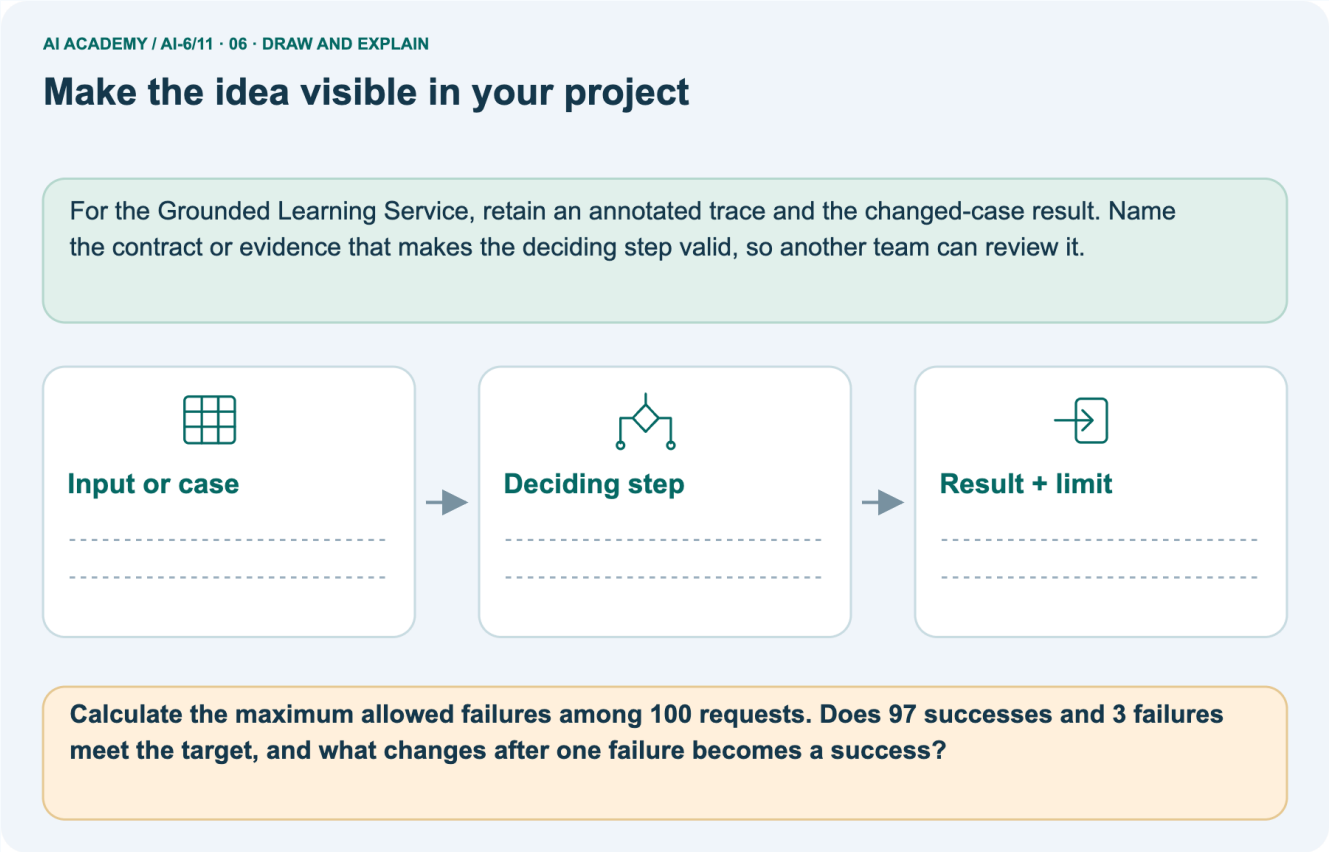
Follow the labelled picture with the matching explanation. Open the vector diagram

An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

Limit: Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.